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High-Temperature CO_2 Sorption on Hydrotalcite Having a High Mg/Al Molar Ratio

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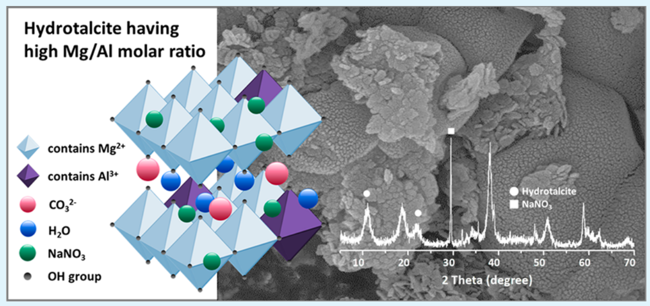
Supporting Information

ABSTRACT: Hydrotalcites having a Mg/Al molar ratio between 3 and 30 have been synthesized as promising high-temperature CO_2 sorbents. The existence of $NaNO_3$ in the hydrotalcite structure, which originates from excess magnesium nitrate in the precursor, markedly increases CO_2 sorption uptake by hydrotalcite up to the record high value of 9.27 mol kg^{-1} at 240 °C and 1 atm CO_2.



KEYWORDS: *hydrotalcite, sorbent, CO_2 sorption, high temperature, Mg/Al ratio*

Because of anthropogenic activities like industrialization during the past century, carbon dioxide (CO_2) emissions and its concentration in the atmosphere have been continuously increasing, resulting in the global warming problem.[1,2] According to the fifth report published by the Intergovernmental Panel on Climate Change (IPCC) in 2013, the amount of greenhouse gas emissions has risen by 2.2% since 2000, and CO_2 accounts for about 78% of these total emissions.[3] The major source of CO_2 emissions is fossil fuel combustion in power plants and energy-intensive industries. Since fossil fuel is still expected to be a major source of energy to meet growing energy needs in the near future, CO_2 capture from flue gases produced by fossil fuel combustion is necessary to help resolve this environmental issue.[4,5]

The temperature of the flue gas stream containing CO_2 typically ranges from 200 to 400 °C, so cooling steps are required to use conventional CO_2 capture methods that are only operational at low temperature. However, to operate power plants and energy intensive processes economically and to capture CO_2 efficiently, direct CO_2 capture from hot flue gases without costly cooling steps is desirable.[6,7] Sorption technology using solid materials having an affinity for high-temperature CO_2 can be applied to direct CO_2 capture in both pre- and post-combustion capture processes.[8,9]

Among various available sorbents, hydrotalcite, also known as layered double hydroxide, is an attractive material for CO_2 capture at high temperature. Hydrotalcites are anionic clay minerals with the formula $[M_{1-x}{}^{2+}M_x{}^{3+}(OH)_2]^{x+}[A_{x/n}{}^{n-}]\cdot mH_2O$, where M^{2+}, M^{3+}, and A^{n-} represent a divalent cation, trivalent cation, and interlayer charge-compensating anion, respectively, where x is commonly in the range of 0.17−0.33.

The hydrotalcite structure consists of positively charged metal-hydroxide layers, in which divalent cations are partially substituted by trivalent cations at the center of octahedral sites. Anions and H_2O molecules reside in the interlayer region, which compensates for the excess positive charge in the hydroxide layer. This arrangement results in a charge-balanced structure.[4,10] In many research studies of hydrotalcite as a CO_2 sorbent, Mg^{2+}, Al^{3+}, and $CO_3{}^{2-}$ have been often used for M^{2+}, M^{3+}, and A^{n-}, respectively.[2,11,12] Mg−Al−CO_3 hydrotalcite has good thermal stability and is regenerable, but its CO_2 sorption capacity is relatively low compared with other sorbents used for high temperature CO_2 sorption.[4,13] Many studies have been conducted to identify factors that can increase the CO_2 sorption capacity of hydrotalcite. Factors including the type of cations or anions used in its synthesis, synthesis method, alkali metal (Na, K) impregnation, support material, and synthesis pH have all been investigated.[2,8,11,14−19]

In this study, the effect of the Mg/Al molar ratio on the CO_2 sorption by hydrotalcite is investigated, with an emphasis on high Mg/Al ratios. According to previous studies the typical Mg/Al molar ratio of hydrotalcite used for CO_2 sorption has been between 2 and 3.[2,10] Although CO_2 sorption on hydrotalcite having higher Mg/Al ratios has been reported, its CO_2 sorption capacity was low and there was no detailed study of higher Mg/Al ratios.[20] We prepared hydrotalcites based on a modified coprecipitation method with Mg/Al ratios between 3

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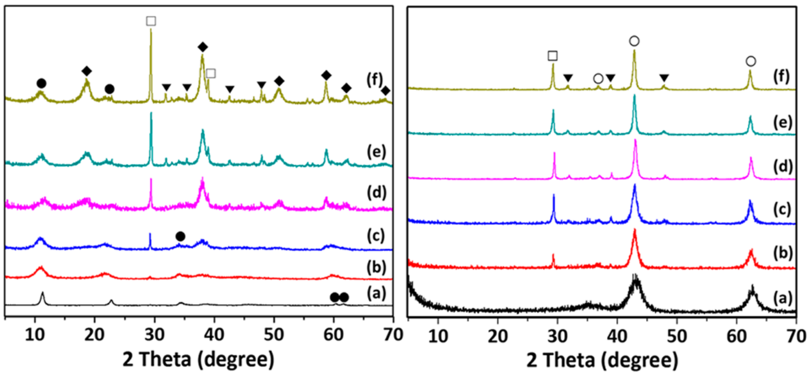

Figure 1. XRD patterns of hydrotalcites with a Mg/Al molar ratio in the feed of (a) 3, (b) 9, (c) 12, (d) 20, (e) 25, and (f) 30. Left, before calcination; right, after calcination. ●, hydrotalcite; ♦, $Mg(OH)_2$; □, $NaNO_3$; ▼, Na_2CO_3; ○, MgO.

and 30. These hydrotalcites were studied to find a specific factor that increases the CO_2 sorption capacity of hydrotalcite in the temperature range of 200−330 °C.

Figure 1 shows the X-ray diffraction (XRD) patterns of hydrotalcites with various Mg/Al molar ratios in the feed. When the Mg/Al ratio is 3, a well-crystallized hydrotalcite structure forms with characteristic peaks at $2\theta \approx 11$, 22, 34, 60, and 61° (JCPDS 70−2151). For hydrotalcites having Mg/Al ratios ranging from 9 to 30, both crystalline hydrotalcite and unreacted $Mg(OH)_2$ phases are detected. The $Mg(OH)_2$ phase is thought to be derived from excess $Mg(NO_3)_2$ in the precursor, which remains after the synthesis of the hydrotalcite structure is completed. As the Mg/Al ratio increases, the hydrotalcite characteristic peaks at $2\theta \approx 11$, 22, and 34° become broader, which implies that the crystallinity of hydrotalcite has decreased. In addition, the characteristic peaks for $Mg(OH)_2$ increase in intensity and become sharper. Excess $Mg(NO_3)_2$, which was originally added as a magnesium precursor, can also be used as a nitrate precursor for the $NaNO_3$ phase in the hydrotalcite structure. Consequently, as the Mg/Al ratio increases from 9 to 30 in the feed, the $NaNO_3$ characteristic peak at $2\theta \approx 29°$ (JCPDS 76−1694) appears and its intensity increases. After calcination at 500 °C, hydrotalcite and $Mg(OH)_2$ phases are transformed into a mixed oxide phase. The $NaNO_3$ peak is still visible because $NaNO_3$ does not decompose at temperatures below 500 °C. Figure S1 shows the weight change of $NaNO_3$ under a flow of N_2 and CO_2 with increasing temperature and implies that the decomposition temperature of $NaNO_3$ is above 600 °C and $NaNO_3$ itself does not sorb CO_2.[21]

N_2 adsorption−desorption isotherms on hydrotalcites with various Mg/Al ratios can be seen in Figure S2. All the samples exhibit type IV isotherm behavior with an H3 hysteresis loop, according to the IUPAC classification. This behavior corresponds to a mesoporous solid with a wide pore size distribution. The pore size distribution calculated by the Barrett−Joyner−Halenda (BJH) method is exhibited in Figure S3. Hydrotalcites having Mg/Al ratios higher than 3 feature a broad pore size distribution. As summarized in Table 1, the Brunauer−Emmett−Teller (BET) surface area and pore volume for hydrotalcite significantly decrease and the relative amount of $NaNO_3$ in hydrotalcite increases as the Mg/Al ratio increases. The large amount of $NaNO_3$ can block or fill the pores in hydrotalcite, resulting in a reduction of the surface area and pore volume.[10,22] In the SEM images of hydrotalcite (Figure S4), aggregated clusters can be seen and sand flower morphology appears when hydrotalcites have high Mg/Al molar ratios of 20 and 30.

Table 1. Textural Properties and CO_2 Sorption Uptake by Hydrotalcites at Various Mg/Al Ratios

Mg/Al Molar Ratio	BET Surface Area ($m^2\ g^{-1}$)	Pore Volume ($m^3\ g^{-1}$)	CO_2 Sorption Uptake ($mol\ kg^{-1}$)[a]	Relative Amount of $NaNO_3$[b]
3	256	0.80	0.83	∼ 0
9	93	0.48	1.22	0.29
12	44	0.19	6.52	1.28
20	33	0.18	9.27	2.36
25	17	0.08	6.56	3.89
30	16	0.07	6.23	5.71

[a] CO_2 sorption was measured by TGA at 240 °C and 1 atm. [b] The relative amount of $NaNO_3$ was estimated by the ratio of the $NaNO_3$ and hydrotalcite peak intensities at their characteristic XRD peak angles $2\theta \approx 29$ and 11°, respectively.

Figure 2 shows the CO_2 sorption behavior of hydrotalcites measured by thermogravimetric analysis (TGA) at 240 °C and 1 atm CO_2. When the Mg/Al molar ratio increases from 3 to 20, the CO_2 uptake at 300 min increases significantly from 0.83 to 9.27 mol kg^{-1}. It is noteworthy that the hydrotalcites with Mg/Al molar ratios between 12 and 30 exhibit totally different two-step CO_2 sorption behaviors compared with the normal single-step CO_2 sorption of the hydrotalcite with a Mg/Al molar ratio of 3. In the hydrotalcites with high Mg/Al ratios,

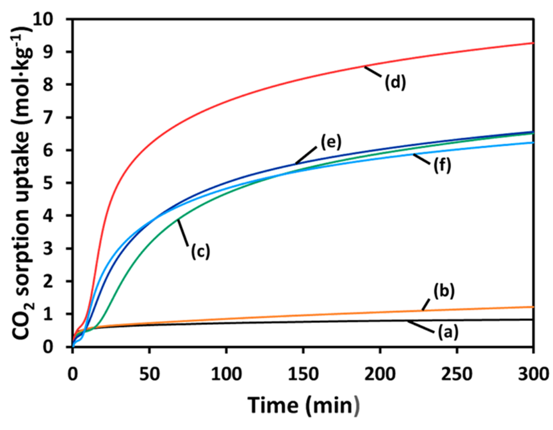

Figure 2. CO_2 sorption behavior at 240 °C and 1 atm CO_2 for hydrotalcite with a Mg/Al molar ratio in the feed of (a) 3, (b) 9, (c) 12, (d) 20, (e) 25, and (f) 30.

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the CO_2 sorption in the first 5−15 min is low, and the initial CO_2 loading is even lower than that of hydrotalcite having a Mg/Al ratio of 3. After the first plateau, the CO_2 sorption goes through a slow transition to a second phase in which CO_2 loading is very high. This drastic increase of CO_2 sorption by hydrotalcites with high Mg/Al ratios has not been previously reported. Interestingly, the considerable increase in CO_2 sorption with Mg/Al ratios up to 20 is in accordance with the sharp increase in the XRD peak intensity of $NaNO_3$ (Figure 1). This result suggests that the existence of $NaNO_3$ in the hydrotalcite structure enhances CO_2 sorption. However, when the Mg/Al ratio is further increased to 25 or 30, the CO_2 sorption uptake decreases slightly. The reason for this decrease is thought to be the loss of pores and reduction of surface area resulting from pore blockage caused by deposition of excess $NaNO_3$. Because of these two conflicting effects of $NaNO_3$, there is an optimal amount of $NaNO_3$ for maximum CO_2 sorption.

In physisorption, sorption uptake normally decreases as the temperature increases, based on exothermic sorption phenomena. When chemisorption is dominant, however, the dependence of sorption uptake on temperature is different.[23,24] To find the effect of temperature on CO_2 sorption, CO_2 sorption tests were carried out using TGA at temperatures ranging from 200 to 330 °C for hydrotalcite having a Mg/Al ratio of 20. Figure 3 shows that CO_2 sorption uptake at 300 min increases as the temperature increases from 200 to 270 °C. The maximum sorption uptake of 9.27 mol kg^{-1} is achieved at 240 °C. However, the sorption uptake sharply decreases to 2.85 and 0.438 mol kg^{-1} at 300 and 330 °C, respectively. The sorption kinetics also vary depending on temperature.

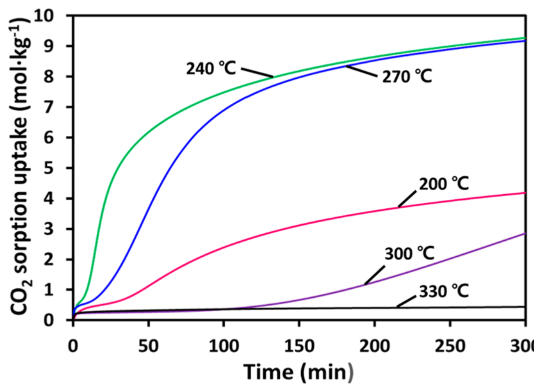

Figure 3. CO_2 sorption behavior of hydrotalcite with a Mg/Al molar ratio of 20 at different temperatures and 1 atm CO_2.

Structural changes of calcined hydrotalcite with a Mg/Al ratio of 20 were studied using in situ XRD analysis over the temperature range of 25 to 500 °C under a flow of N_2 and CO_2 (Figure 4). Under both N_2 and CO_2 flows, sharp MgO peaks at $2\theta \approx 42$ and 62° (JCPDS 77−2179) are detected for the main structure. The intensity of the characteristic peak for $NaNO_3$ at $2\theta \approx 29°$ decreases as the temperature increases above 240 °C, and disappears at temperatures above 300 °C. Above the melting temperature of 308 °C, the crystalline phase of $NaNO_3$ is completely transformed into an amorphous phase.[25] Compared with the XRD spectra under N_2 flow, new peaks corresponding to $MgCO_3$ appear at $2\theta \approx 33$ and 53° (JCPDS 86−2347) under CO_2 flow between 240 and 400 °C. This result shows that some CO_2 combines with MgO to form

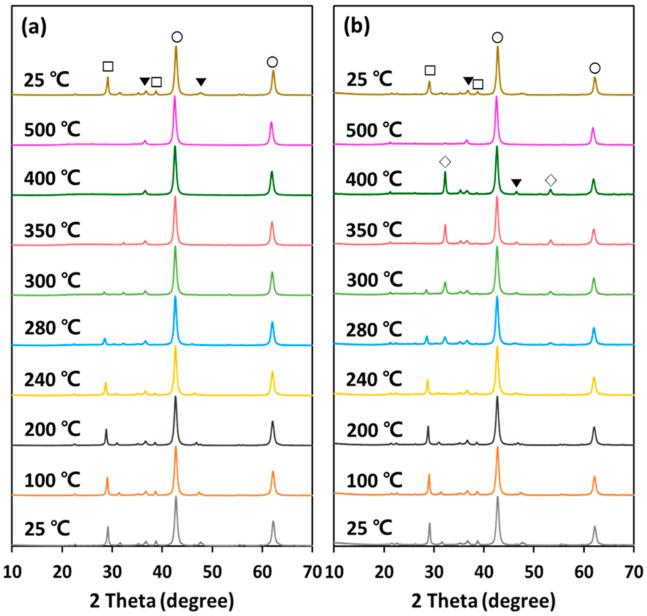

Figure 4. In situ XRD patterns of calcined hydrotalcite with a Mg/Al molar ratio of 20 at different temperatures under a flow of (a) N_2 and (b) CO_2. ○, MgO; ◊, $MgCO_3$; □, $NaNO_3$; ▼, Na_2CO_3.

$MgCO_3$ in hydrotalcite. As the temperature increases, the $MgCO_3$ peak intensity increases, which implies a greater amount of CO_2 was sorbed in the form of $MgCO_3$. From this result, CO_2 sorption on hydrotalcite is considered to be caused by the reaction to form $MgCO_3$. However, CO_2 sorption is also affected by adsorption since the reduction in CO_2 sorption uptake appeared when surface area and pore volume of hydrotalcite decreased in the case of Mg/Al ratio of 25 and 30 (Table 1). Therefore, CO_2 loading on hydrotalcites having a high Mg/Al ratio is thought to be based on both adsorption and reaction. At 500 °C, the characteristic peaks of $MgCO_3$ disappear due to the decomposition of $MgCO_3$. The same XRD patterns are obtained for CO_2 and N_2 flows, suggesting that hydrotalcite can be completely regenerated at 500 °C. After cooling to 25 °C, the XRD spectra obtained before heating are regenerated, which confirms the thermal reconstruction of the initial structure.

Harada et al. and Zhang et al. reported MgO-based CO_2 sorbents promoted with alkali metal nitrate and inferred that $NaNO_3$ on the surface of solid MgO enhances the rapid generation of active MgO sites and carbonate ions required for the formation of $MgCO_3$ crystals.[26,27] Mixed oxide, Mg(Al)O, is considered an active site for CO_2 sorption on hydrotalcite and this structure is highly similar to that of MgO.[2] Therefore, it is thought that the crystallization of $MgCO_3$ on hydrotalcites having a high Mg/Al ratio is promoted by the existence of $NaNO_3$ in a similar manner to the MgO-based CO_2 sorbent.

To further examine the control and effect of the amount of $NaNO_3$ in hydrotalcite, we compared three different washing conditions during the preparation of hydrotalcite with a Mg/Al ratio of 12: Vacuum filtration of the aged solution (1) without additional washing, (2) with washing using 100 mL of distilled water (unless otherwise specifically stated, samples were synthesized under this condition), and (3) washing with 250 mL of distilled water. $NaNO_3$ is soluble in water, and therefore the remaining amount of $NaNO_3$ in hydrotalcite can be controlled by varying the amount of water used in the washing

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step.[25] The XRD patterns of the hydrotalcite samples (Figure S5) show that the amount of $NaNO_3$ decreases as the amount of water used in the washing step increases. When the hydrotalcite is washed with 250 mL of water, the $NaNO_3$ peak is not present. In addition, CO_2 sorption uptake by the hydrotalcites prepared using the different washing conditions was evaluated by TGA, as shown in Figure S6. The highest sorption uptake of 6.52 mol kg^{-1} is obtained when hydrotalcite is washed with 100 mL water, and the CO_2 sorption uptake drastically decreases to 0.48 mol kg^{-1} for the hydrotalcite washed with 250 mL of water, in which little $NaNO_3$ remains. The plot of CO_2 sorption uptake versus relative content of $NaNO_3$ in hydrotalcite (Figure S7) clearly shows that the amount of $NaNO_3$ is a key factor for enhanced CO_2 sorption by hydrotalcite, and that there exists an optimal amount of $NaNO_3$ for the highest CO_2 uptake.

A cyclic sorption and desorption test was carried out for hydrotalcite having a Mg/Al ratio of 20, as shown in Figure S8. The change in CO_2 uptake was recorded over 16 cycles of CO_2 sorption at 240 °C for 4 h under a flow of CO_2 and desorption at 400 °C for 1 h under a flow of N_2. The CO_2 uptake noticeably reduced during the first 5 cycles and then became stabilized with slight decrease. This implies that the CO_2-sorbed hydrotalcite can be regenerated with a temperature and pressure swing sorption scheme. Although the hydrotalcite having high Mg/Al ratio maintained only 25% of the initial CO_2 sorption uptake after 16 cycles, its working capacity is still high compared to that of other hydrotalcites. The CO_2 sorption working capacity is expected to be enhanced after the optimization of sorption and desorption procedure.

Figure S9 shows CO_2 sorption equilibrium isotherm for hydrotalcite having a Mg/Al ratio of 20. The data was measured by using TGA at 240 °C with changing CO_2 partial pressure in the range of 0.05−1 atm. As CO_2 partial pressure increased, CO_2 sorption uptake on hydrotalcite also increased and approached a limiting value.

In summary, Mg−Al−CO_3 hydrotalcites with Mg/Al molar ratios between 3 and 30 were prepared from nitrate-based precursors by a modified coprecipitation method and the effect of Mg/Al molar ratio on the characteristics and CO_2 sorption of hydrotalcite was studied. XRD analysis exhibits the existence of $NaNO_3$ in the synthesized hydrotalcite, which was produced from the excessive nitrate ions originating from the magnesium nitrate precursor. The CO_2 sorption uptake on hydrotalcite having a Mg/Al molar ratio of 20 was 9.27 mol kg^{-1} at 240 °C, which is the highest value so far reported. In addition, when the Mg/Al molar ratio was between 12 and 30, hydrotalcites showed a unique two-step CO_2 sorption behavior: low CO_2 loading initially followed by a gradual transition to very high CO_2 loading. The results show that the quantity of $NaNO_3$ is closely related to the CO_2 sorption of hydrotalcite and the existence of $NaNO_3$ promotes the CO_2 sorption. The in situ XRD result also suggests that CO_2 is sorbed to form $MgCO_3$ in the temperature range between 240 and 400 °C, and hydrotalcite can be completely regenerated at 500 °C.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acsami.5b12598.

Additional TGA, N_2 adsorption analysis, SEM images, XRD, and CO_2 sorption results (PDF)

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Notes

The authors declare no competing financial interest.

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